16/10/2026 Photonic Qubits (lec 3)

Review: photons, photonic encodings
BS, DV vs CV
2-mode transformations
HoM interference
CNOT in linear optics
fusion-based quantum computation

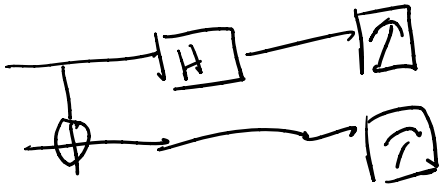
① BSM – teleportation, entanglement swapping, fusion-based Q. comp

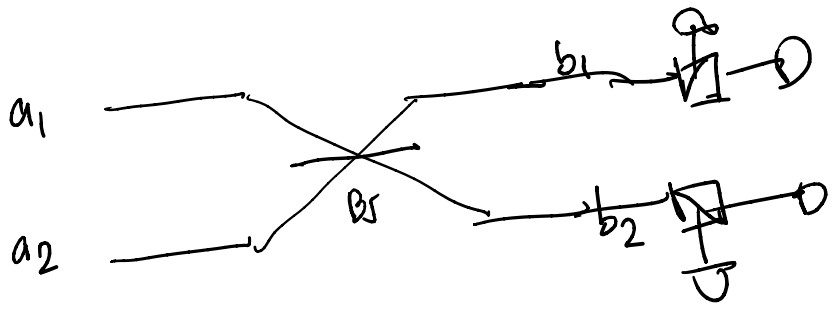
② CNOT, CX that uses entanglement

③ 2 maps M_1, M_2

④ physical implementation of CX gate

① BSM - Bell state measurement





Unitary for BS: $\begin{pmatrix} b_1 \\ b_2 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$

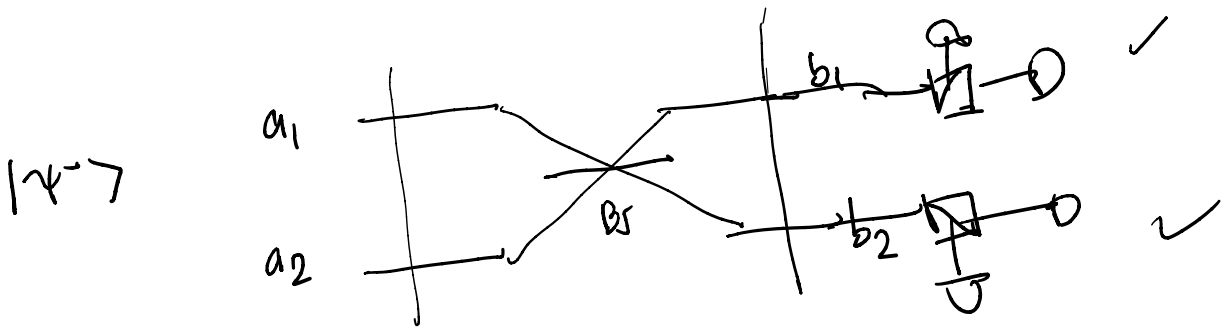
Unitary for BS: $\begin{pmatrix} b_1 \\ b_2 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$

$b = U a$

$a = U^\dagger b$ ✓

$\begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}$

BSM - distinguishes 4 Bell states



✓ ✓

$$|\psi^-\rangle = \frac{1}{\sqrt{2}} \left(|HV\rangle - |VH\rangle \right)$$

$$|\psi^-\rangle = \frac{1}{\sqrt{2}} \left(a_{1H} a_{2V} - a_{1V} a_{2H} \right) |0\rangle$$

$$= \frac{1}{\sqrt{2}} \left[\frac{1}{2} (b_{1H} - b_{2H})(b_{1V} + b_{2V}) - \frac{1}{2} (b_{1V} - b_{2V})(b_{1H} + b_{2H}) \right]$$

$$= \frac{1}{2\sqrt{2}} \Big[b_{1H} b_{1V} - b_{2H} b_{2V} - b_{2H} b_{1V} + b_{1H} b_{2V} - (b_{1V} b_{1H} - b_{2V} b_{2H} - b_{2V} b_{1H} + b_{1V} b_{2H}) \Big]$$

$$- b_{1V} b_{1H} + b_{2V} b_{2H} + b_{2V} b_{1H} - b_{1V} b_{2H}$$

$$= \frac{1}{2\sqrt{2}} \left\{ 2 b_{2V} b_{1H} - 2 b_{2H} b_{1V} \right\}$$

$$= \frac{1}{\sqrt{2}} \left(|HV\rangle - |VH\rangle \right) = |\psi^-\rangle !$$

eigenstate of BS!

Other Bell states:

(1) → 2 photons, (2) → 0 photons

$$|\psi^+\rangle \rightarrow \frac{1}{\sqrt{2}}\left(|HV,0\rangle + |0,HV\rangle\right)$$

$$|\Phi^+\rangle \rightarrow \frac{1}{2}\left(|HH,0\rangle - |0,HH\rangle + |VV,0\rangle - |0,VV\rangle\right)$$

$$|\Phi^-\rangle \rightarrow \frac{1}{2}\left(|HH,0\rangle - |0,HH\rangle - |VV,0\rangle + |0,VV\rangle\right)$$

cannot be distinguished

$|\psi^+\rangle$ & $|\psi^-\rangle$ can be detected = 50% success rate

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CX control $c=0$ – target doesn't change

target $c=1$ – target flips

an ideal CX creates entanglement between c & t

$$|\psi\rangle_c = |+\rangle \quad , \quad |\psi\rangle_t = |H\rangle$$

$$\left(\frac{|H\rangle_c + |V\rangle_c}{\sqrt{2}}\right)|H\rangle_t = \frac{|HH\rangle_{ct} + |VV\rangle_{ct}}{\sqrt{2}}$$

CX that uses entanglement to create the entanglement between c & t

Inputs: $|\psi\rangle_c = \alpha|H\rangle + \beta|V\rangle$

$$|\psi\rangle_t = \gamma|D\rangle + \delta|A\rangle$$

$$= \gamma\frac{(|H\rangle+|V\rangle)}{\sqrt{2}} + \delta\frac{(|H\rangle-|V\rangle)}{\sqrt{2}}$$

$$= \frac{1}{\sqrt{2}}\left((\gamma+\delta)|H\rangle + (\gamma-\delta)|V\rangle\right)$$

$$|\psi\rangle_c \otimes |\psi\rangle_t = |\psi\rangle_{ct}$$

$$|\psi\rangle_{ct} = \frac{1}{\sqrt{2}}\left[\alpha(\gamma+\delta)|HH\rangle + \alpha(\gamma-\delta)|HV\rangle + \beta(\gamma+\delta)|VH\rangle + \beta(\gamma-\delta)|VV\rangle\right]$$

Consider M_1, M_2

$$M_1 = |H\rangle\langle H|_c \otimes |H\rangle\langle H|_t + |V\rangle\langle V|_c \otimes |V\rangle\langle V|_t$$

$$M_2 = |D\rangle\langle D|_t + |A\rangle\langle A|_t$$

$M_2 M_1 |\psi_c\rangle |\Phi^+\rangle |\psi_t\rangle$ = ideal CX

$M_1 |\psi_c\rangle |\Phi^+\rangle = \left[|H\rangle\langle HH| + |V\rangle\langle VV| \right] \left(\alpha |H\rangle_c + \beta |V\rangle_c \right)$

$\left(\frac{1}{\sqrt{2}} |HD\rangle_{12} + |HA\rangle_{12} + |VD\rangle_{12} - |VA\rangle_{12} \right)$

$= \alpha |H\rangle\langle H| + \beta |V\rangle\langle V| \qquad \left[\frac{1}{\sqrt{2}} \left(|HD\rangle_{12} + |HA\rangle_{12} + |VD\rangle_{12} - |VA\rangle_{12} \right) \right]$

$= \alpha |H\rangle_c |D\rangle_2 + \alpha |H\rangle_c |A\rangle_2 + \beta |V\rangle_c |D\rangle_2 - \beta |V\rangle_c |A\rangle_2$

$M_2 M_1 |\psi_c\rangle |\Phi^+\rangle |\psi\rangle_t$

$= \left(|D\rangle_t\langle DD|_{2t} + |A\rangle_t\langle AA|_{2t} \right) \left(\alpha |H\rangle_c |D\rangle_2 + \alpha |H\rangle_c |A\rangle_2 + \beta |V\rangle_c |D\rangle_2 - \beta |V\rangle_c |A\rangle_2 \right) \left(\gamma |D\rangle + \delta |A\rangle \right)$

$= \left(\gamma |D\rangle_t\langle D|_2 + \delta |A\rangle_t\langle A|_2 \right) (\quad)$

$= \alpha\gamma |HD\rangle_{ct} + \beta\gamma |VD\rangle_{ct} + \alpha\delta |HA\rangle_{ct} - \beta\delta |VA\rangle_{ct}$

$\left(\frac{1}{\sqrt{2}} |H\rangle_t + |V\rangle_t \right) \quad \frac{1}{\sqrt{2}} |H\rangle_t + |V\rangle_t \quad \frac{1}{\sqrt{2}} (|H\rangle_t - |V\rangle_t) \quad \frac{1}{\sqrt{2}} (|H\rangle_t - |V\rangle_t)$

$= \frac{1}{\sqrt{2}} \left\{ \alpha\gamma \left(|HH\rangle + |HV\rangle \right) + \beta\gamma \left(|VH\rangle + |VV\rangle \right) + \alpha\delta \left(|HH\rangle - |HV\rangle \right) - \beta\delta \left(|VH\rangle - |VV\rangle \right) \right\}$

$= \frac{1}{\sqrt{2}} \left\{ \alpha(\gamma+\delta) |HH\rangle + \alpha(\gamma-\delta) |HV\rangle + \beta(\gamma-\delta) |VH\rangle + \beta(\gamma+\delta) |VV\rangle \right\}$

compare w/ the original $|\psi\rangle_{ct}$

$|\psi_{ct}\rangle = \frac{1}{\sqrt{2}} \left\{ \alpha(\gamma+\delta) |HH\rangle + \alpha(\gamma-\delta) |HV\rangle + \beta(\gamma+\delta) |VH\rangle + \beta(\gamma-\delta) |VV\rangle \right\}$

$VV \to VH$ & $VH \to VV$

CX gate

How to implement experimentally?

PBS:



$\alpha|HH\rangle + \beta|HV\rangle + \gamma|VH\rangle + \delta|VV\rangle$

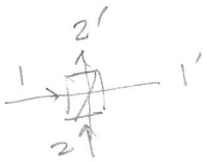
$\rightarrow \alpha|H,H\rangle + \beta|HV,0\rangle + \gamma|0,VH\rangle + \delta|V,V\rangle$

failure

failure

if polarization input is the same, there is always one photon in each output port

How to implement M_1 & M_2



$$M_1 = |H\rangle\langle HH| + |V\rangle\langle VV|$$

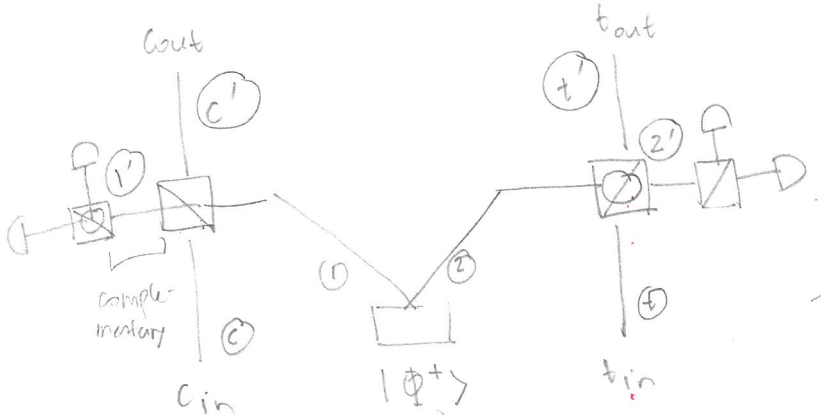
PBS:

$$\alpha|HH\rangle + \beta|HV\rangle + \gamma|VH\rangle + \delta|VV\rangle$$

$$\rightarrow \alpha|H,H\rangle + \beta|HV,0\rangle + \gamma|0,VH\rangle + \delta|V,V\rangle$$

($|HV,0\rangle$: failure; $|0,VH\rangle$: failure)

- post selecting on one port ensures there is a photon of the same polarisation in the other port
- if polarisation is the same, there is always one photon in each output port



$\Phi^+ \sim |HV\rangle + |VH\rangle$

$|H\rangle = |0\rangle$

$|V\rangle = |1\rangle$

1', 2' } ancillary

If control is in superposition $|+\rangle_c |H\rangle_t$

the CX creates entanglement between c & t

left PBS w/ one photon in c' & 1' are $|H_c H_A\rangle, |V_c V_A\rangle$

↓

Z-basis parity check

for arbitrary control $f|H\rangle + g|V\rangle$

we need to measure in complementary basis $f|H_c H_B\rangle \pm g|V_c V_B\rangle$

right PBS

↓

X-basis parity check

each PBS checks parity, but will succeed only $\frac{1}{2}$

so total success is $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$

M_1 & M_2 are the basis for the fusion operators

↓ type 1 ↓ type 2